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Assignment 6

Introduction

In this lab, I worked on creating a mobile application using Flutter that can receive push notifications. I used Firebase Cloud Messaging (FCM) to enable real-time communication between the server and my application. The main goal was to understand how notifications are sent and received on a mobile device.

Objective

The objective of this lab was to build a Flutter application that can request notification permissions, receive push notifications, and display them both on the screen and inside the app.

Steps I Followed

First, I created a Flutter application and connected it to Firebase. After that, I integrated Firebase Cloud Messaging into the app. I configured the app to request notification permission from the user and ensured that it can receive messages.

I then retrieved the device token and displayed it in the console. Using this token, I went to the Firebase Console and sent a test notification to my device. Finally, I tested the app in both foreground and background modes.

Implementation

I implemented notification handling so that messages can be received and displayed in different states of the application. When a notification arrives, it is shown in the notification bar and also inside the app interface. I also added a popup to alert the user when a message is received.

Results

The application successfully received notifications sent from Firebase. I was able to see the message content in the notification bar and inside the app. The popup feature also worked correctly when a notification was received.

Observations

I observed that Firebase Cloud Messaging is reliable and easy to integrate with Flutter. The device token plays an important role in sending notifications to a specific device. I also noticed

that handling notifications differs depending on whether the app is in the foreground or background.

Challenges

One of the challenges I faced was configuring permissions correctly, especially for different devices. Another challenge was handling notifications when the app is open, as it requires additional implementation compared to background notifications.

Conclusion

Through this lab, I learned how to integrate Firebase Cloud Messaging into a Flutter application. I now understand how push notifications work and how to handle them effectively in a mobile app.